Weather-Based Prediction of Wind Turbine Energy Output

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1. Introduction

- **Project Title:** Weather-Based Prediction of Wind Turbine Energy Output: A Next-Generation Approach to Renewable Energy Management
- **Team Members:**
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  - Dake Vijay Kumar
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2. Project Overview

- **Purpose:**
  The purpose of this project is to build a machine learning–based system that predicts wind turbine energy output using historical turbine data and live weather inputs.
- **Features:**
  - Data preprocessing and cleaning.
  - Random Forest regression model for prediction.
  - Flask-based web dashboard for user interaction.
  - Integration with OpenWeather API for real-time weather data.
  - Visualization of actual vs predicted power outputs.

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3. Architecture

- **Frontend:**
  Flask templates (HTML, CSS, JavaScript) used for UI design and dashboard visualization.
- **Backend:**
  Python Flask application handling API requests, ML model predictions, and weather data integration.
- **Database:**
  Local CSV dataset (T1.csv) for training and testing. Model stored as .sav file using Joblib. Future scope includes cloud database integration (MongoDB Atlas / AWS RDS).

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4. Setup Instructions

- Prerequisites:
  - Python 3.9+
  - Flask
  - Pandas, NumPy, Scikit-learn, Matplotlib
  - Joblib
  - OpenWeather API key
- Installation:

1. Clone the repository.
2. Install dependencies using pip install -r requirements.txt.
3. Set up environment variables (API key for OpenWeather).
4. Run the Flask server with python app.py.

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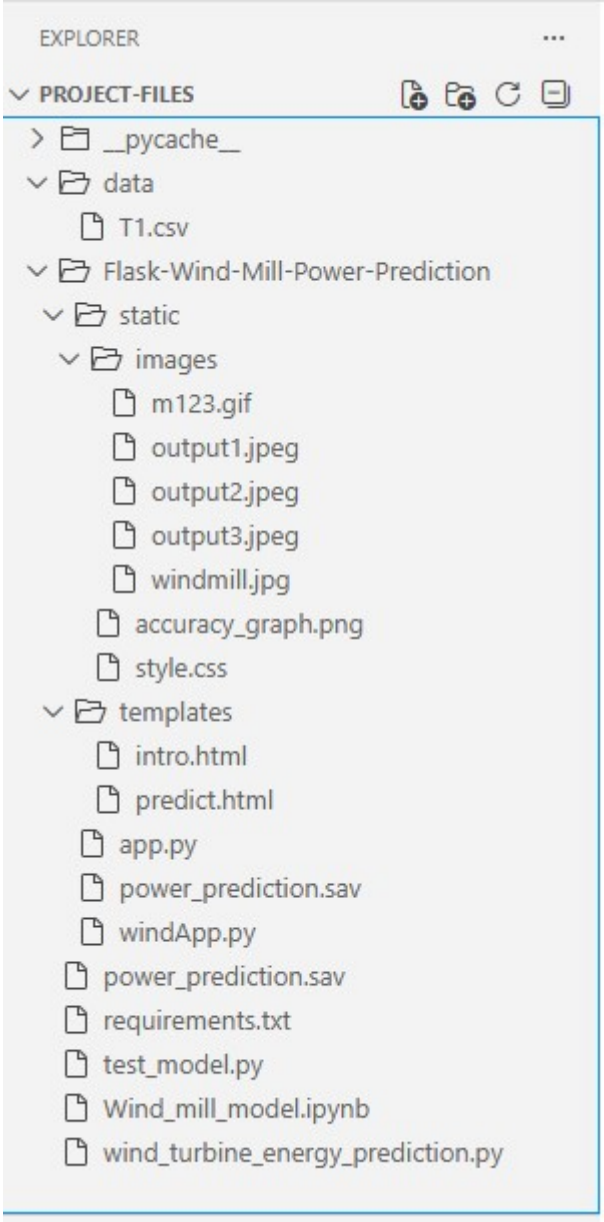
5. Folder Structure

- Client (Frontend):
  - templates/ → HTML files (intro page, dashboard).
  - static/ → CSS, JS, and images.
- Server (Backend):
  - app.py → Flask application.
  - model_training.py → ML model training script.
  - power_prediction.sav → Saved Random Forest model.
  - data/ → Dataset files.



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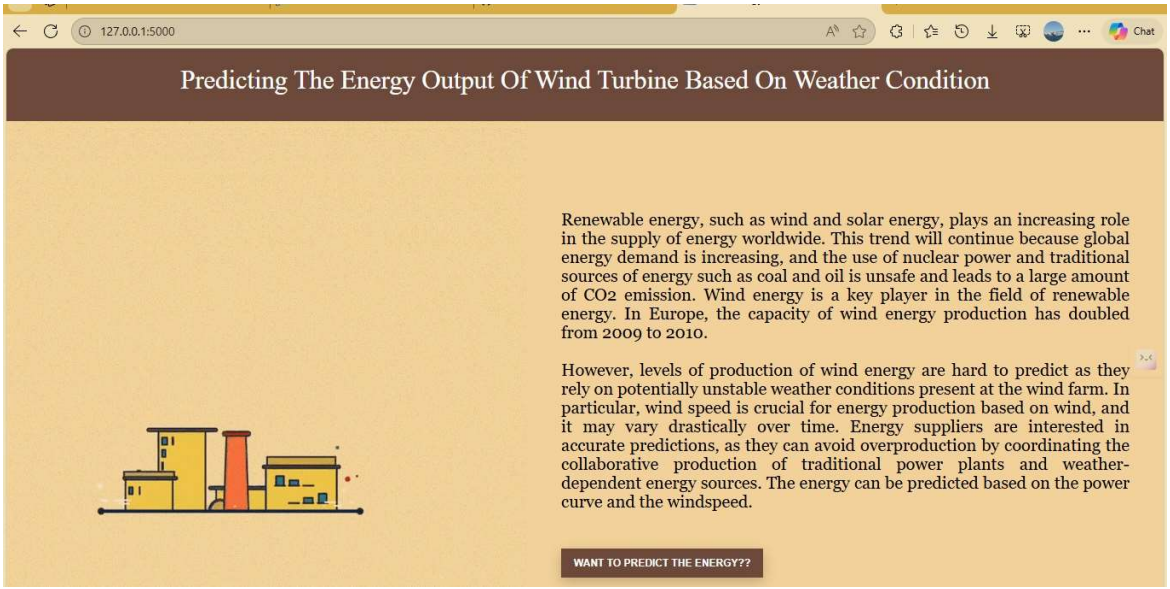
6. Running the Application

- Frontend: Runs automatically via Flask templates.
- Backend: Start with python app.py.
- Access at http://127.0.0.1:5000/.



7. API Documentation

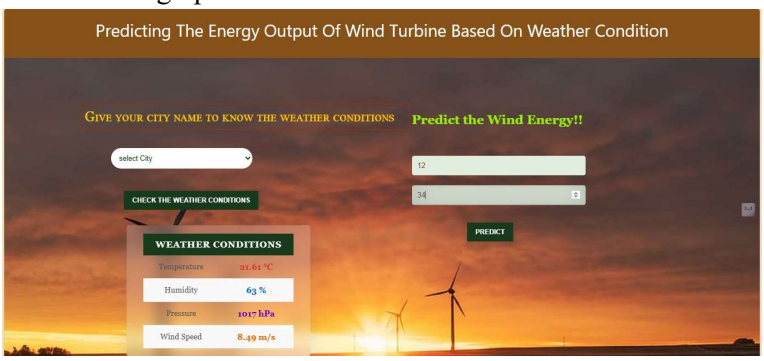
- Endpoint 1: /predict
  - Method: POST
  - Parameters: TheoreticalPower, WindSpeed
  - Response: Predicted Active Power
- Endpoint 2: /weather
  - Method: GET
  - Parameters: City name
  - Response: Weather data (temperature, humidity, pressure, wind speed)

8. Authentication

- Currently open access.
- API key required for OpenWeather API integration.
- Future scope: JWT-based authentication for secure access.

9. User Interface

- Intro page with project overview.
- Dashboard with:
  - Weather data display.
  - Prediction module.
  - Visualization graphs.



10. Testing

- Unit testing for ML model predictions.
- API testing for weather data retrieval.
- Functional testing for input validation and dashboard navigation.

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; C:\Users\LENOVO\Desktop\Final_year_projects\APSCHE-Project\NEW1\Flask-Wind-Mill-Power-Prediction> python app.py
 \Users\LENOVO\AppData\Roaming\Python\Python313\site-packages\sklearn\base.py:442: InconsistentVersionWarning: Trying to unpickle estimator DecisionTreeRegressc
 version 1.8.0 when using version 1.7.2. This might lead to breaking code or invalid results. Use at your own risk. For more info please refer to:
:tps://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
 warnings.warn(
 \Users\LENOVO\AppData\Roaming\Python\Python313\site-packages\sklearn\base.py:442: InconsistentVersionWarning: Trying to unpickle estimator RandomForestRegressc
 version 1.8.0 when using version 1.7.2. This might lead to breaking code or invalid results. Use at your own risk. For more info please refer to:
:tps://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
 warnings.warn(
 Serving Flask app 'app'
 Debug mode: off
RNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 Running on http://127.0.0.1:5000
ess CTRL+C to quit
```

11. Screenshots or Demo

- Scatter plot of actual vs predicted power.
- Dashboard screenshot showing weather + prediction results.
- Line chart trends of wind speed vs power output.

12. Known Issues

- Limited dataset size may affect generalization.
- API errors if invalid city names are entered.
- No authentication for dashboard access (future enhancement needed).

13. Future Enhancements

- Deploy on cloud (AWS/GCP/Azure).
- Add JWT authentication for secure access.
- Expand dataset for improved accuracy.
- Add more visualizations (heatmaps, time-series forecasting).
- Integrate grid demand APIs for real-time energy balancing.

14. Conclusion

The **Wind Turbine Energy Prediction project** successfully demonstrates the application of machine learning and data-driven techniques to solve a real-world renewable energy

challenge. By leveraging historical turbine data, preprocessing methods, and a **Random Forest regression model**, the project achieves strong predictive accuracy, enabling stakeholders to forecast energy output with confidence.

The integration of a **Flask-based dashboard** and **OpenWeather API** ensures that predictions are not only technically sound but also accessible and user-friendly. Visualizations such as scatter plots, line charts, and correlation heatmaps further enhance interpretability, making the solution practical for energy companies, wind farm operators, and grid managers.

Through structured **Agile sprint planning, backlog management, and performance testing**, the project was executed with clarity and measurable progress. Testing confirmed the robustness of the model, while defect analysis and bug tracking highlighted areas for improvement.

Ultimately, this project provides a scalable foundation for future enhancements, including cloud deployment, advanced authentication, expanded datasets, and integration with grid demand APIs. It stands as a strong example of combining **data science, software engineering, and agile methodology** to deliver a solution that addresses both technical and customer-centric needs in the renewable energy sector.